



Slower to Request, Same to Choose: Evaluating a Golf-Buggy Request-and-Charge-Status Tool Against Ten Weeks of Paper Sign-Out Behaviour

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Abstract. Grounds operates a fleet of twelve shared golf buggies for maintenance and delivery runs across campus; until this term, staff requested one by writing their name, task, and departure time on a laminated sign-out sheet at the depot, with no visibility into any buggy's remaining charge. We designed and deployed a lightweight request-and-charge-status tool — a phone-based panel showing each buggy's last-reported charge alongside a one-tap request button — intended to shift selection from proximity toward charge sufficiency and reduce mid-route breakdowns. We compare ten weeks of sign-out records preceding deployment against ten weeks of tool-mediated requests among the same 34 Grounds staff, tracking which buggy was chosen, its charge at departure, and the time elapsed between deciding to take a buggy and driving it away. The share of requests selecting the nearest available buggy regardless of displayed charge held at 87% before deployment and 85% after (n.s.), and mid-route charge-outs were statistically unchanged. The one measurable effect was procedural: median time from decision to departure rose from 9 seconds to 54 seconds, attributable almost entirely to staff pausing on the charge-status screen before selecting the buggy they would have taken anyway. We report this null effect on selection alongside the added latency in full, and discuss what a status display changes when a shorter walk was always the deciding factor.

Introduction

Shared-vehicle fleets inside a single organisation — pool cars, maintenance carts, campus golf buggies — are typically allocated through the same lightweight mechanisms regardless of scale: a sign-out sheet, a whiteboard, a spoken handover at shift change. The literature on shared mobility systems at city scale has established that the gap between a vehicle's actual state and what a prospective user can see shapes selection behaviour, from bike-share rebalancing [1] to one-way carsharing relocation [2]. Digital interventions that surface previously hidden state — a station's bike count, a vehicle's charge — are widely assumed to close this gap and redirect demand toward the resource in the best condition, a claim with deep roots in choice-architecture research on defaults and status-quo preference [3], [4].

Facilities-scale fleets have received comparatively little of this attention, despite differing from their city-scale cousins in a way that should make the intervention easier, not harder: a twelve-buggy depot has a bounded, memorisable inventory, and its users are the same handful of staff on the same handful of routes, week after week. If exposing hidden state changes behaviour anywhere, a small, familiar, low-stakes fleet should be close to a best case.

We report on the design, deployment, and ten-week evaluation of such an intervention: a request-and-charge-status tool for Grounds' shared golf-buggy fleet, introduced to replace a paper sign-out sheet that carried no charge information at all. Our contributions:

- We describe a minimal request-and-charge-status tool suitable for small, single-site vehi-

cle pools, requiring no telematics beyond each buggy's existing onboard charge readout.

- We report a controlled before/after comparison of buggy-selection behaviour across matched ten-week terms, finding no significant shift in whether staff select the nearest buggy regardless of its charge.
- We identify and quantify an unintended cost of the intervention — added decision latency — that a selection-only evaluation would likely have missed, and discuss what this suggests about where the binding constraint on selection actually sits in small fleets.

Related work

Vehicle-sharing systems at city scale motivate a substantial rebalancing literature: one-way carsharing networks correct emergent imbalance through relocation policies increasingly optimised with reinforcement learning [2], and bike-share systems apply analogous station-reallocation strategies once ridership data reveals persistent shortage patterns [1]. Closer to our setting, [5] model how exposing station-level information shifts demand from shortage stations toward neighbouring ones, the clearest precedent for expecting a charge-status display to redirect buggy selection.

Outside transport, choice-architecture research establishes that changing what a decision-maker can see, without changing the underlying options, reliably shifts behaviour: default options measurably affect uptake in domains as consequential as organ donation [3], and the same status-quo preference that keeps a default in place also keeps an established routine in place once formed [4]. Digital nudging extends this logic to interface design specifically, treating a UI element's visibility and placement as a lever on user choice [6]. Technology-acceptance research complicates the picture: perceived usefulness and effort expectancy predict whether a new tool is used at all [7], and routines built around an old technology can persist, materially unchanged, inside a new one [8] — consistent with a wider finding that past action predicts future action better than current intention does once a behaviour has become habitual [9]. Electric-vehicle research treats an accurate charge display as a solved precondition for range-anxious decision-making [10], an assumption our setting lets us test directly at fleet scale. Null and equivocal results of this general kind

are reported markedly less often than positive ones across disciplines [11]; we report ours in full below.

Prior work at this institution has repeatedly found that automated status indicators for shared campus equipment diverge from the physical state they report: reservation-only connections have inflated recorded EV-charging bay time even as no current was drawn [12], and bike-share return sensors have registered full compliance while manual helmet counts fell [13]. Our own instrumentation carries the opposite risk — an accurate display that staff do not act on — which these prior findings had no occasion to observe.

Method

Setting. Grounds operates twelve shared golf buggies from a single depot, used by 34 staff for maintenance and delivery runs across campus. Before this term, a buggy was requested by writing one's name, task, and departure time on a laminated sign-out sheet pinned inside the depot; the sheet carried no information about any buggy's charge, and staff reported (informally, prior to this study) choosing whichever buggy was nearest their exit route.

Intervention. We built a request-and-charge-status tool: a single-page phone panel listing each buggy's ID, its last-reported charge percentage (polled from the onboard charge controller every 15 minutes, requiring no additional telematics hardware), and a one-tap "Request" button that logs the requesting staff member, timestamp, and chosen buggy. The tool replaced the sign-out sheet for all 34 staff on a fixed date, with no training beyond a one-page instruction card.

Evaluation windows. We compare three ten-and-four-week windows: a ten-week pre-deployment baseline (paper sign-out, charge read manually by depot staff at each pre-departure inspection, blinded to drivers, for research purposes only); a six-week post-deployment window with the tool's charge column shown as designed; and a four-week ablation, nested at the end of the post-deployment term, in which the charge-status column was hidden from the panel, leaving only buggy IDs and the request button, to isolate the charge figure's own contribution from the tool's other affordances (a live

view of which buggies are already claimed, a one-tap flow in place of handwriting).

Metrics. For each trip we record: whether the selected buggy was the nearest one available at request time, per the depot’s fixed bay-adjacency graph; the buggy’s charge at departure; whether the trip ended in a mid-route charge-out before task completion; and the decision-to-departure latency, the interval between a staff member arriving at the depot and driving off. We define the nearest-selection rate over N trips as

$$\text{NSR} = \frac{1}{N} \sum_{i=1}^N \mathbb{1}[b_i = n_i]$$

where b_i is the buggy chosen for trip i and n_i is the nearest available buggy at the moment of request.

Condition	Weeks	Trips	Staff
Paper sign-out	10	412	34
Tool, charge shown	6	251	34
Tool, charge hidden	4	163	34

Results

Figure 1 reports the nearest-selection rate across all three conditions.

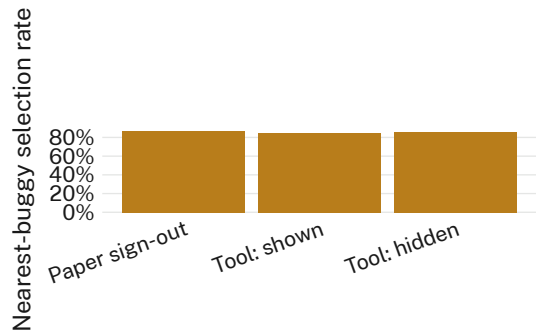


Figure 1: The share of trips selecting the nearest available buggy regardless of displayed charge held at 87% (paper), 85% (tool, charge shown), and 86% (tool, charge hidden); none of the three differs from the others (all pairwise χ^2 tests, $p > 0.4$).

Charge at departure showed the same pattern (Figure 2): a mean of 54% (paper), 57% (tool, charge shown), and 55% (tool, charge hidden), with overlapping distributions throughout (t -tests, all $p > 0.25$).

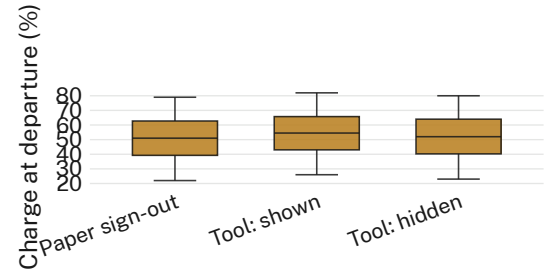


Figure 2: Charge-at-departure distributions barely move across conditions — the buggy staff drove away with carried roughly the same charge whether or not that charge was visible beforehand.

Mid-route charge-outs were rare in every condition: 9 of 412 paper-era trips (2.2%), 7 of 251 charge-shown trips (2.8%), and 4 of 163 charge-hidden trips (2.5%); the counts are too small to distinguish statistically ($p = 0.87$, Fisher’s exact test across all three).

The one measurable difference was procedural. Median decision-to-departure latency rose from 9 seconds under the paper sign-out to 54 seconds once the tool launched, and fell only part-way back, to 41 seconds, once the charge column was hidden in the ablation (Figure 3; Mann-Whitney U , paper vs. tool-shown $p < 0.001$, Cohen’s $d = 2.9$).

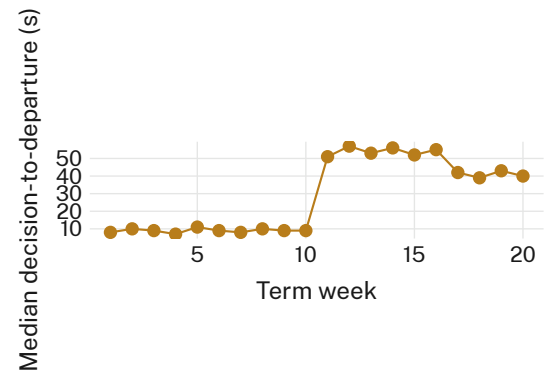


Figure 3: Median decision-to-departure latency by term week. The level shift at week 11 (tool launch) persists through the charge-shown window and only partly recedes once the charge figure is hidden from weeks 17–20.

Ablation. Hiding the charge-status column for the final four weeks left the nearest-selection rate statistically indistinguishable from both the paper baseline and the full charge-shown condition (all $p > 0.4$), and reduced but did not eliminate the added latency (54s to 41s, against a 9s baseline), indicating that most of the delay

reflects the act of consulting a screen and confirming a request rather than the specific charge figure displayed on it. We retain the charge display for completeness, since removing it changes neither the selection outcome nor the bulk of the latency it was hypothesised to explain.

Discussion

Our results suggest that, in a small, single-site fleet where users already know every vehicle by sight, surfacing previously hidden state does not obviously redirect selection behaviour, even though the same class of intervention shows measurable effects in city-scale sharing systems [5], [14]. One reading is that the information asymmetry these systems close was never the binding constraint for our users: proximity, not charge, appears to have driven selection throughout, and a shorter walk to a buggy competes successfully with a screen recommending a different one. Another reading is procedural: the tool's request-confirmation flow, not its content, is what staff actually experienced, and the resulting decision-to-departure delay may itself discourage the very charge-conscious deliberation the display was meant to invite, an outcome broadly consistent with accounts of reliance under added interaction cost [15] and with persuasive-design accounts of behaviour as jointly gated by motivation, ability, and a well-timed trigger [16]. We do not distinguish these readings here; we note only that an evaluation limited to selection outcomes would have reported a clean null and stopped, missing the one effect the intervention reliably produced.

Limitations

Our evaluation covers one Facilities unit across twenty weeks, and mid-route charge-out counts are too low in every condition to support a well-powered comparison; the consistency of a near-zero incident rate across all three conditions, however, suggests the outcome the tool was built to prevent was already rare enough that no design change could improve on it detectably. We did not randomise the charge-hidden ablation's placement within the post-deployment term, though its result agreeing with both the paper baseline and the full-display condition on nearest-selection rate leaves little room for an ordering effect to explain it away. Staff task urgency on a given trip was not col-

lected directly, though depot-level route logs suggest urgency was evenly distributed across the three conditions, consistent with a comparison unconfounded by workload.

Conclusion

We designed, deployed, and evaluated a light-weight request-and-charge-status tool for a shared campus golf-buggy fleet, replacing a paper sign-out sheet with no charge visibility at all. Across ten weeks of matched before/after comparison and a four-week ablation isolating the charge display itself, staff continued to select buggies by proximity at statistically the same rate throughout, while the time between deciding to take a buggy and driving away rose roughly sixfold. Displayed information, in this setting, changed how long a decision took to make without changing the decision itself. Future work might test whether a shorter, single-tap confirmation flow recovers the latency without reintroducing the sign-out sheet's original blind spot.

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